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(1)

Coupled Pendulum

Date of experiment: **Partner's name:**

Date of Lab. report submission:

Aim of the Experiment

To determine the resonance and beat time period of a coupled pendulum.

Apparatus Required

(i) Coupled pendulum set up (connected to computer and printer).

(ii)Stop watch.

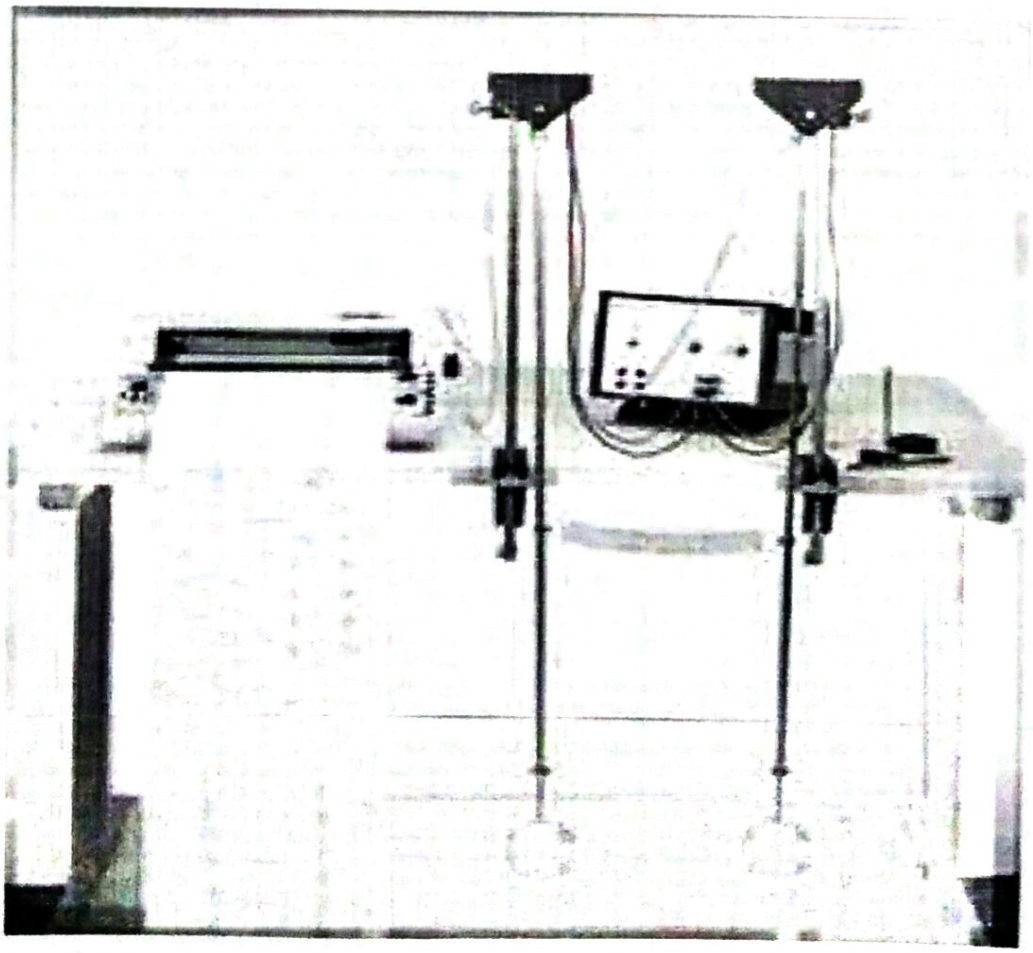

Fig-1: Experimental set up

Observations

Table-1: Resonance and beat time period.

UNCOUPLED		COUPLED							
Time for 20 oscillations	Average T	IN-PHASE		OUT OF PHASE		BEAT		RESONANCE	
		Time for 20 Oscillations		Time for 20 Oscillations		Time for 5 stops in secs		Time for 5 Oscillations in secs	
t_1= 40.68	Avg. T=	t_1= 40.06	Avg. T_0=	t_1= 35.59	Avg. T_1=	t_1= 115.48	Avg. T_b= CT_b=Beat Time period	t_1= 9.86	T_c= (Resonance Time period)
t_2= 41.02	2.0323 ±0.005	t_2= 40.16	2.0043 ±0.005	t_2= 36.80	1.8193 ±0.005	t_2= 112.58	~~5.7050 ±0.005~~ 22.82 ±0.002	t_2= 9.92	1.9820 ±0.002
t_3= 40.24		t_3= 40.04		t_3= 36.77		t_3= 114.24		t_3= 9.95	

Calculations :

Estimated time period for beat :

$$T_b = 2\frac{T_0 T_1}{T_0 - T_1} = \left(\frac{2 \times 2.0043 \times 1.8193}{2.0043 - 1.8193}\right) s$$

$$= 39.4208 \text{ s}$$

Estimated time period for resonance :

$$T_c = 2\frac{T_0 T_1}{T_0 + T_1} = \left(\frac{2 \times 2.0043 \times 1.8193}{2.0043 + 1.8193}\right) s$$

$$= 1.9073 \text{ s}$$

χ degree of coupling :-

$$\chi = \frac{\omega_1^2 - \omega_0^2}{\omega_1^2 + \omega_0^2} = \frac{T_0^2 - T_1^2}{T_0^2 + T_1^2}$$

$$= \frac{(2.0043)^2 - (1.8193)^2}{(2.0043)^2 + (1.8193)^2}$$

$$= 0.0965 .$$

Error in T_c :-

Observed T_c (s)	Estimated T_c (s)	$\overline{T_c}$ (s)	$\sigma^2 = \frac{\sum d_i^2}{n}$	σ	Error % $= \frac{\sigma}{\overline{T_c}} \times 100$
1.9820	1.9073	1.9447	0.00139	0.0372	1.91 %

$$\% \text{ Error} = \frac{\sigma}{\overline{T_c}} \times 100\ \%$$

$$= \left(\frac{0.0372}{1.9447} \times 100\right)\%$$

$$= 1.91\ \%$$

Error in T_b :

Observed T_b (s)	Estimated T_b (s)	$\overline{T_b}$ (s)	$\sigma^2 = \frac{\sum d_i^2}{n}$	σ	% Error $= \sigma / \overline{T_b} \times 100$
22.82	39.4208	31.1204	34.445	5.869	18.86 %

$$\% \text{ Error} = \frac{\sigma}{\overline{T_b}} \times 100\ \%$$

$$= \left(\frac{5.869}{\cancel{78.8}\ 31.1204} \times 100\right)\%$$

$$= 18.86\ \%$$

Conclusion:

1. Through observation, resonance time period obtained is 1.982 ± 0.002 (s).

2. Through calculation, resonance time period so obtained is 1.9073 ± 0.0013 (s).

3. Through observation, half beat time period is 22.82 ± 0.002 (s).

4. Through calculation, half beat time period is 39.4208 ± 5.86 (s)

5. Degree of coupling is 0.0965.

6. This experiment proves the fact that the theory of coupled oscillators can be used to model various other oscillating system such as atoms and nuclei and radiation and absorption of energy.